

Aliasing Error Estimation of Fractional Fourier Transform Based on Windowed Samples*

Fang-Jia Yan¹, Bing-Zhao Li^{2*},

¹School of Mathematics and Statistics, Beijing Institute of Technology, Beijing, China

²Beijing Key Laboratory on MCAACI, Beijing Institute of Technology, Beijing, China

*Email address: li_bingzhao@bit.edu.cn

The fractional powers of the Fourier transform operator is a generalization of the Fourier transform. Although there are many works on the derivation of sampling about fractional Fourier transform, studies on the aliasing error are rare. This paper presents a new aliasing error estimation method of fractional Fourier transform connecting with Poisson sum formula based on windowed samples.

Index Terms—Aliasing error estimate, Fractional Fourier transform, Windowed sampling

I. INTRODUCTION

Fractional Fourier transform (FRFT) is a parameterised transform with an adjustable transform parameter which makes it more flexible and superior over classical Fourier transform (FT) in several applications especially in nonstationary signals processing. So the fractional powers is a variation of standard FT [1]. Therefore, many properties of FRFT, such as its convolution, Poisson sum formula and sampling theorems can be derived from those of FT by a series of mathematical transformations [2]–[4].

Sampling expansion of FRFT from a set of samples or measurements is one of the fundamental direction in signal processing.

The spectrum leakage takes place during spectrum analysis due to finite length of discrete FT outside selected interval. So as to control spectrum leakage, functions that satisfying finite length and zero-valued outside selected interval, are known as window function like Hann-window and Blackmann-window which are special cases of traditional windows in the form of a sum of cosine terms [5]–[7], they are regularly used in digital signal processing, digital filter design, spectrum estimation and other applications.

Because there always exists diverse kinds of errors that might affect the accuracy of the reconstruction. Hence, the study of error estimation is strongly significant in sampling community. In the periodic FT case, aliasing error which takes no account of truncation for non-continuous function is estimated [8]. In FT domain, the signal function truncated

by the window function will behave leakage, and the truncated product is no longer band-limited unless it is a null function. Truncation error and aliasing error estimation in Shannon sampling theorem caused by above condition based on window sampling have been derived [9]. Based on the sampling theorem in fractional wavelet transform which is applied to recover a signal, errors arise from how many items are needed to calculate so that the recovered signal is close to the original one as expected have been estimated [10]. Fixed-point error in computing the discrete fractional Fourier transform (DFRFT) is analyzed [11]. In FRFT domain, prolate spheroidal wave functions are used to sample a time-limited signal, then estimates the upper bound of the aliasing error and reduces the aliasing error of the recovered signal [12]. Truncation error about a new proposed sampling theorem without band-limiting constraints is analyzed [13]. In linear canonical transform domain, aliasing error and truncation error for uniformly sampling a random signal are estimated [14].

The purpose of this paper is to analyze aliasing error of fractional Fourier transform based on windowed samples. The structure of paper is as follows. First we review basic knowledge about FRFT, such as the relationship between FT and FRFT, frequency shifting property, convolution, sampling theorem and Poisson summation formula. Then we introduce the window function and norms used in this paper. Finally, based on aliasing phenomenon in windowed sampling, the error is expressed.

II. PRELIMINARIES

A. Backgrounds of FRFT

From the perspective of linear integral transformation, the FRFT of a signal function $f(t)$ with order p is defined as [3].

$$\mathcal{F}_p[f](u) = \int_{-\infty}^{\infty} \tilde{K}_p(u, t) f(t) dt \quad (1)$$

where

$$\tilde{K}_p(u, t) =$$

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$$\begin{cases} A_\alpha \exp\{j\frac{t^2}{2} \cot(\alpha) - jut \csc(\alpha) + j\frac{u^2}{2} \cot(\alpha)\}, \\ \alpha \neq n\pi \\ \delta(t-u), \alpha = 2n\pi \\ \delta(t+u), \alpha = 2n\pi - \pi \end{cases} \quad (2)$$

is the kernel function of FRFT. The constants are

$$A_\alpha = \sqrt{\frac{1-j\cot(\alpha)}{2\pi}}, \alpha = \frac{p\pi}{2}, 0 < |p| < 2, 0 < |\alpha| < \pi$$

A signal function $f(t)$ is band-limited in FRFT domain when

$$\mathcal{F}_p[f](u) = 0 \text{ for } |u| > \Omega_\alpha \quad (3)$$

where Ω_α is the bandwidth of signal $f(t)$.

In this paper, we assume $\alpha > 0$ for convenience and use the transform when $\alpha \neq n\pi$ to analyze more general situation. Based on the mentioned case, the definition of DFRFT is [3]

$$\begin{aligned} & \mathcal{F}_p[f]^*(u) \\ &= \Delta t \sum_{k=-\infty}^{\infty} A_\alpha f(k\Delta t) \\ & \times \exp\{j\frac{(k\Delta t)^2}{2} \cot(\alpha) - juk\Delta t \csc(\alpha) + j\frac{u^2}{2} \cot(\alpha)\} \end{aligned} \quad (4)$$

It is well known that the FRFT degenerate into standard FT when $p = 1$ and the one with $p = 0$ corresponds to the identity operator. Therefore, the following relationship between FT and FRFT is obtained through some integral transformations and variable substitutions.

Property 1. Let $f^\wedge(\omega)$ be the Fourier transform of $f(t)$, and $\mathcal{F}_p[f](u)$ be the p order FRFT. Set $h(t) = \exp\{j\frac{t^2}{2} \cot(\alpha)\} f(t)$. The FRFT of $f(t)$ is equivalent to the FT of $h(t)$ [4].

$$\mathcal{F}_p[f](u) \exp\{-j\frac{t^2}{2} \cot(\alpha)\} = A_\alpha h^\wedge(u \csc(\alpha)) \quad (5)$$

Let $v = u \csc(\alpha)$. We have

$$\mathcal{F}_p[f](v \sin(\alpha)) \exp\{-j\frac{(v \sin(\alpha))^2}{2} \cot(\alpha)\} = A_\alpha h^\wedge(v) \quad (6)$$

Property 2. The FRFT of a signal function $s(t)$ multiplying a frequency shifting function is [3]

$$\begin{aligned} & \mathcal{F}_p[s(t) \exp\{jtv\}](u) \\ &= \mathcal{F}_p[s(t)](u - v \sin(\alpha)) \\ & \times \exp\{juv \cos(\alpha) - j\frac{v^2}{2} \sin(\alpha) \cos(\alpha)\} \end{aligned} \quad (7)$$

Property 3. The convolution equation of FRFT is [15]

$$\begin{aligned} & \mathcal{F}_p[f(t)g(t) \exp\{j\frac{t^2}{2} \cot(\alpha)\}](u) = \\ & \sqrt{\frac{1+j\cot(\alpha)}{2\pi}} \exp\{j\frac{u^2}{2} \cot(\alpha)\} (\widetilde{\mathcal{F}}_p[f] * \widetilde{\mathcal{F}}_p[g])(u) \end{aligned} \quad (8)$$

where

$$\widetilde{\mathcal{F}}_p[f](x) = \mathcal{F}_p[f](x) \exp\{-j\frac{x^2}{2} \cot(\alpha)\} \quad (9)$$

and

$$(f * g)(x) = \int_{-\infty}^{\infty} f(t)g(x-t)dt \quad (10)$$

Property 4. Set sampling frequency $\Omega_s = \frac{2\pi}{\Delta t}$, sampling theorem for band-limited signal $f(t)$ in FRFT states that [3]

$$\Omega_s \geq 2\Omega_\alpha |\csc(\alpha)| \quad (11)$$

B. The Poisson summation formula

It is well known that the Poisson summation formula can be used for Shannon sampling theorem proving. In Fourier domain, the meaning of the Poisson summation formula is that the time domain periodic extension is equivalent to the frequency domain sampling, which can be represented as follows from mathematical perspective with interval Δt . In FT domain,

$$\sum_{k=-\infty}^{\infty} f(t+k\Delta t) = \frac{1}{\Delta t} \sum_{n=-\infty}^{\infty} f^\wedge\left(\frac{n}{\Delta t}\right) \exp\{j\frac{n}{\Delta t}t\} \quad (12)$$

or

$$\sum_{k=-\infty}^{\infty} f(k\Delta t) = \frac{1}{\Delta t} \sum_{n=-\infty}^{\infty} f^\wedge\left(\frac{n}{\Delta t}\right) \quad (13)$$

In FRFT domain, the Poisson summation formula associated with FRFT can be derived as [4]

$$\begin{aligned} & \sum_{k=-\infty}^{\infty} f(t+k\Delta t) \exp\{j\frac{(t+k\Delta t)^2}{2} \cot(\alpha)\} \\ &= \frac{1}{A_\alpha \Delta t} \sum_{n=-\infty}^{\infty} \mathcal{F}_p[f]\left(\frac{n}{\Delta t} \sin(\alpha)\right) \\ & \times \exp\{-j\frac{(n \sin(\alpha)/\Delta t)^2}{2} \cot(\alpha) + j\frac{n}{\Delta t}t\} \end{aligned} \quad (14)$$

or

$$\begin{aligned} & \sum_{k=-\infty}^{\infty} f(k\Delta t) \exp\{j\frac{(k\Delta t)^2}{2} \cot(\alpha)\} \\ &= \frac{1}{A_\alpha \Delta t} \sum_{n=-\infty}^{\infty} \mathcal{F}_p[f]\left(\frac{n}{\Delta t} \sin(\alpha)\right) \\ & \times \exp\{-j\frac{(n \sin(\alpha)/\Delta t)^2}{2} \cot(\alpha)\} \end{aligned} \quad (15)$$

Proof. Combining (6) and (12), one verifies that (14) and (15) are easy consequences of simple calculations. $\square$

III. WINDOW FUNCTION AND NORMS

The window functions are weighting functions which are applied to sampling. The product that window multiply with signal function is done to transform an infinitely long function to finite in length so that the frequency spectrum of a signal can be measured. Window functions often have

following common properties like real, even, non-negative and time-limited, also their Fourier transforms have main lobe at the origin and side lobes at both sides. To reduce sampling error, the selection of an appropriate window function appears to be especially important. In order to facilitate the error estimation of the next section, the concept of norm used is introduced in following.

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined on the whole line. The 1-norm of f is defined to be $\|f\|_1 = \int_{-\infty}^{\infty} |f(t)|dt$. If f is defined on a closed interval $[a, b]$, the 1-norm and ∞ -norm are defined as

$$\begin{aligned}\|f\|_{1,[a,b]} &= \int_a^b |f(t)|dt \\ \|f\|_{\infty,[a,b]} &= \sup_{t \in [a,b]} |f(t)|\end{aligned}\quad (16)$$

In this paper, window functions $g_{\mathbb{R}}$ that used in sampling satisfies two conditions

- $g_{\mathbb{R}}(t)$ is compact on $[-R, R]$;
- $g_{\mathbb{R}}(t) = g(t/R)$, g is an even function and compact on $[-1, 1]$.

IV. ALIASING ERROR ANALYZE IN FRFT

In this section, we derive aliasing error in FRFT based on basic theory and common sense.

The root cause of the aliasing phenomenon is sampling [9]. If the sampling cannot collect enough information of the original signal, aliasing will occur. The frequency aliasing phenomenon is high and low frequency components are confused due to changes in the spectrum of the sampled signal. If sampling frequency is not high enough, the sampled points which represent the sample values of the low-frequency signals and represent the sample values of the high-frequency signals as well. When the signal is reconstructed, the high-frequency signals are replaced by low-frequency signals, and two waveforms overlap completely that lead to severe distortion.

In FRFT domain, the sampling error can be written as:

$$|\mathcal{F}_p[f](u) - \mathcal{F}_p[f]^*(u)| \quad (17)$$

In order to calculate aliasing error, we substitute $\mathcal{F}_p[f \cdot g_{\mathbb{R}}]^*(u)$ for $\mathcal{F}_p[f]^*(u)$, then (17) can be rewritten as

$$\begin{aligned}& |\mathcal{F}_p[f](u) - \mathcal{F}_p[f \cdot g_{\mathbb{R}}]^*(u)| \\ & \leq \underbrace{|\mathcal{F}_p[f](u) - \mathcal{F}_p[f \cdot g_{\mathbb{R}}](u)|}_{\text{truncation error}} \\ & \quad + \underbrace{|\mathcal{F}_p[f \cdot g_{\mathbb{R}}](u) - \mathcal{F}_p[f \cdot g_{\mathbb{R}}]^*(u)|}_{\text{aliasing error}}\end{aligned}\quad (18)$$

Next, we estimate the aliasing error in (18) and which can be represented as the following theorem 1.

Theorem 1 (aliasing error).

$$\begin{aligned}& \mathcal{F}_p[f \cdot g_{\mathbb{R}}](u) - \mathcal{F}_p[f \cdot g_{\mathbb{R}}]^*(u) \\ & = - \sum_{k \in \mathbb{Z}/\{0\}} \mathcal{F}_p[f \cdot g] \left(\frac{k \sin(\alpha)}{\Delta t} + u \right) \\ & \quad \times \exp \left\{ j \left(\frac{ku \cos(\alpha)}{\Delta t} + \frac{\cot(\alpha)}{2} \left(\frac{k \sin(\alpha)}{\Delta t} \right)^2 \right) \right\}\end{aligned}\quad (19)$$

Proof. By definition we have

$$\begin{aligned}& \mathcal{F}_p[f \cdot g_{\mathbb{R}}]^* \\ & = A_{\alpha} \sum_{-\infty}^{+\infty} \exp \left\{ j \frac{(k\Delta t)^2 + u^2}{2} \cot(\alpha) - jk\Delta tu \csc(\alpha) \right\} \\ & \quad \times f(k\Delta t) g_{\mathbb{R}}(k\Delta t) \Delta t \\ & = A_{\alpha} \Delta t \exp \left\{ j \frac{\cot(\alpha)}{2} u^2 \right\} \\ & \quad \times \sum_{-\infty}^{+\infty} (f(k\Delta t) g_{\mathbb{R}}(k\Delta t) \exp \{-ju \csc(\alpha) k\Delta t\}) \\ & \quad \times \exp \left\{ j \frac{(k\Delta t)^2}{2} \cot(\alpha) \right\}\end{aligned}\quad (20)$$

Now we can substitute (15) into (20), i.e.

$$\begin{aligned}& = \sum_{-\infty}^{+\infty} \mathcal{F}_p[f(\cdot) g_{\mathbb{R}}(\cdot) \exp \{-ju \csc(\alpha)(\cdot)\}] \left(\frac{k \sin(\alpha)}{\Delta t} \right) \\ & \quad \times \exp \left\{ -j \frac{\cot(\alpha)}{2} \left(\frac{k \sin(\alpha)}{\Delta t} \right)^2 \right\} \\ & \quad \times \exp \left\{ j \frac{\cot(\alpha)}{2} u^2 \right\}\end{aligned}\quad (21)$$

Then, substitute (7) into (21), we obtain

$$\begin{aligned}& = \sum_{-\infty}^{+\infty} \mathcal{F}_p[f \cdot g_{\mathbb{R}}] \left(\frac{k \sin(\alpha)}{\Delta t} + u \right) \\ & \quad \times \exp \left\{ -\frac{1}{2} j (u \csc(\alpha))^2 \sin(\alpha) \cos(\alpha) \right\} \\ & \quad \times \exp \left\{ -ju \csc(\alpha) \frac{k \sin(\alpha)}{\Delta t} \cos(\alpha) \right\} \\ & \quad \times \exp \left\{ -j \frac{\cot(\alpha)}{2} \left(\frac{k \sin(\alpha)}{\Delta t} \right)^2 \right\} \\ & \quad \times \exp \left\{ j \frac{\cot(\alpha)}{2} u^2 \right\} \\ & = \sum_{-\infty}^{+\infty} \mathcal{F}_p[f \cdot g_{\mathbb{R}}] \left(\frac{k \sin(\alpha)}{\Delta t} + u \right)\end{aligned}$$

$$\times \exp \left\{ -j \left(\frac{ku \sin(\alpha)}{\Delta t} + \frac{\cot(\alpha)}{2} \left(\frac{k \sin(\alpha)}{\Delta t} \right)^2 \right) \right\} \quad (22)$$

□

When we set $k = 0$, then (22) = $\mathcal{F}_p[f \cdot g_R](u)$, we get

$$\begin{aligned} & \mathcal{F}_p[f \cdot g_R](u) - \mathcal{F}_p[f \cdot g_R]^*(u) \\ &= \mathcal{F}_p[f \cdot g_R](u) - \sum_{-\infty}^{+\infty} \mathcal{F}_p[f \cdot g_R] \left(\frac{k \sin(\alpha)}{\Delta t} + u \right) \\ & \times \exp \left\{ -j \left(\frac{ku \sin(\alpha)}{\Delta t} + \frac{\cot(\alpha)}{2} \left(\frac{k \sin(\alpha)}{\Delta t} \right)^2 \right) \right\} \quad (23) \\ &= - \sum_{k \in \mathbb{Z} \setminus \{0\}} \mathcal{F}_p[f \cdot g] \left(\frac{k \sin(\alpha)}{\Delta t} + u \right) \\ & \times \exp \left\{ j \left(\frac{ku \cos(\alpha)}{\Delta t} + \frac{\cot(\alpha)}{2} \left(\frac{k \sin(\alpha)}{\Delta t} \right)^2 \right) \right\} \end{aligned}$$

Theorem 2 (aliasing error estimation). *Let Δt fulfill the condition $\Omega_\alpha \leq \frac{\pi}{\Delta t \lceil \csc(\alpha) \rceil}$. Then for each $0 < \Omega_1 < \Omega_\alpha$ and $u \in [-\Omega_1, \Omega_1]$, we have*

$$\begin{aligned} & |\mathcal{F}_p[f \cdot \hat{g}_R](u) - \mathcal{F}_p[f \cdot \hat{g}_R]^*(u)| \\ & \leq A_\alpha \|f^\wedge\|_{\infty, [-\Omega_\alpha, \Omega_\alpha]} \\ & \times \|g_R^\wedge\|_{1, \mathbb{R} \setminus [u - \frac{\pi}{\Delta t \lceil \csc(\alpha) \rceil}, u + \frac{\pi}{\Delta t \lceil \csc(\alpha) \rceil}]} \quad (24) \end{aligned}$$

where $\hat{g}_R(t) = g_R(t) \exp\{j \frac{t^2}{2} \cot(\alpha)\}$.

Proof. By theorem 1, we have

$$\begin{aligned} & |\mathcal{F}_p[f \cdot \hat{g}_R](u) - \mathcal{F}_p[f \cdot \hat{g}_R]^*(u)| \\ & \leq \sum_{k \in \mathbb{Z} \setminus \{0\}} \left| \mathcal{F}_p[f \cdot \hat{g}_R](u + \frac{k \sin(\alpha)}{\Delta t}) \right| \quad (25) \end{aligned}$$

By equation (8), the right hand of (25) is

$$\begin{aligned} & \sum_{k \in \mathbb{Z} \setminus \{0\}} \left| \mathcal{F}_p[f \cdot \hat{g}_R](u + \frac{k \sin(\alpha)}{\Delta t}) \right| \\ &= \sum_{k \in \mathbb{Z} \setminus \{0\}} A_\alpha \left| (\tilde{\mathcal{F}}_p[f] * \tilde{\mathcal{F}}_p[g_R])(u + \frac{k \sin(\alpha)}{\Delta t}) \right| \quad (26) \end{aligned}$$

Therefore, for any $u \in [-\Omega_1, \Omega_1]$,

$$\begin{aligned} & |\mathcal{F}_p[f \cdot \hat{g}_R](u) - \mathcal{F}_p[f \cdot \hat{g}_R]^*(u)| \\ & \leq A_\alpha \sum_{k \in \mathbb{Z} \setminus \{0\}} \left| \int_{-\Omega_\alpha}^{\Omega_\alpha} \tilde{\mathcal{F}}_p[f](z) \tilde{\mathcal{F}}_p[g_R](u + \frac{k \sin(\alpha)}{\Delta t} - z) dz \right| \\ & \leq A_\alpha \sum_{k \in \mathbb{Z} \setminus \{0\}} \int_{-\Omega_\alpha}^{\Omega_\alpha} \left| \tilde{\mathcal{F}}_p[f](z) \tilde{\mathcal{F}}_p[g_R](u + \frac{k \sin(\alpha)}{\Delta t} - z) \right| dz \quad (27) \end{aligned}$$

Note that $|\tilde{\mathcal{F}}_p[f](z)| = |f^\wedge(z)|$,

$$\begin{aligned} & |\mathcal{F}_p[f \cdot \hat{g}_R](u) - \mathcal{F}_p[f \cdot \hat{g}_R]^*(u)| \\ & \leq A_\alpha \sum_{k \in \mathbb{Z} \setminus \{0\}} \int_{-\Omega_\alpha}^{\Omega_\alpha} \left| f^\wedge(z) g_R^\wedge(u + \frac{k \sin(\alpha)}{\Delta t} - z) \right| dz \\ & \leq A_\alpha \|f^\wedge\|_{\infty, [-\Omega_\alpha, \Omega_\alpha]} \\ & \times \sum_{k \in \mathbb{Z} \setminus \{0\}} \|g_R^\wedge\|_{1, [u + (2k-1) \frac{\pi}{\Delta t \lceil \csc(\alpha) \rceil}, u + (2k+1) \frac{\pi}{\Delta t \lceil \csc(\alpha) \rceil}]} \\ & = A_\alpha \|f^\wedge\|_{\infty, [-\Omega_\alpha, \Omega_\alpha]} \|g_R^\wedge\|_{1, \mathbb{R} \setminus [u - \frac{\pi}{\Delta t \lceil \csc(\alpha) \rceil}, u + \frac{\pi}{\Delta t \lceil \csc(\alpha) \rceil}]} \quad (28) \end{aligned}$$

□

V. CONCLUSION

In this paper, starting from the definition of FRFT, we first reviewed some important properties of it. Then, based on window sampling, we derived aliasing error associated with FRFT. In our future work, precise aliasing error limitation with other valid conditions will be carried out and other sampling errors in FRFT such as truncation, amplitude and jitter errors will be investigated.

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